

HHS Health Sector AI RFI — Public Comment Submission

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Docket: HHS RFI - Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care" (90 FR 60108, December 23, 2025)

Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The most consequential barrier to AI adoption in clinical care is not regulatory burden, reimbursement lag, or technical immaturity. It is the absence of governance infrastructure that would make adoption safe to accelerate.

Private sector AI innovators face a structural paradox: the clinical environments they are selling into cannot answer three foundational questions about the AI systems already operating within them. Who is specifically responsible for a clinical AI system's outputs — not who is aware of it, but whose name is attached to the outcome? How do we know if it is working — is there a metric, attached to a threshold, that someone can point to? And what triggers when it isn't — when that metric crosses a threshold, what is structurally required to happen?

The evidence for the consequences of this absence is specific. In a 2025 survey, the American Medical Association found that 61% of physicians reported AI is increasing prior authorization denials, 82% reported patients abandoning treatment as a result of AI-driven prior authorization, and 94% reported negative clinical outcomes associated with AI-mediated decision-making.¹ These are not speculative risks. They are outcomes already occurring in clinical environments where AI systems operate between physician and patient without accountability architecture connecting outputs to responsible parties.

The barrier extends beyond clinical settings into institutional procurement. Across industries, 75% of organizations have adopted AI usage policies, but only 36% have implemented formal governance frameworks, and only 48% actively monitor production AI systems for accuracy, drift, or misuse — a figure that drops to 9% among smaller organizations.² Only 24% of organizations can control AI agent actions with guardrails and live monitoring.³ These figures are directional — drawn from surveyed respondents across industries — and are corroborated by higher-authority sources: GAO's federal AI audits, Cisco's readiness index, and CSA/Google's governance maturity correlation data all point to the same structural gap. Where these conditions obtain in health systems, adoption

stalls — not from resistance, but from the entirely rational recognition by procurement, legal, and clinical leadership that deploying AI into a space where accountability, measurement, and consequence are absent creates liability without governance.

The result is a market failure of a specific kind: AI vendors capable of delivering genuine clinical value cannot cross the procurement threshold because the institutional buyer lacks the governance architecture to accept accountability for what the system does. This is not fixed by reducing the regulatory burden on vendors. It is fixed by helping clinical institutions build the accountability infrastructure that makes them capable of adopting AI responsibly. Evidence from the enterprise sector supports this: organizations with comprehensive AI governance were nearly twice as likely to report successful early adoption of agentic AI (46%) compared to organizations with governance still in development (12%).⁴ This pattern holds across clinical AI categories — from EHR-integrated clinical decision support and in-ICU prediction tools to prior authorization systems — wherever accountability architecture is absent at deployment.

¹ American Medical Association, "How AI Is Leading to More Prior Authorization Denials," 2025. <https://www.ama-assn.org/practice-management/prior-authorization/how-ai-leading-more-prior-authorization-denials>

² Pacific AI / Gradient Flow, "2025 AI Governance Survey," December 2025.

³ Cisco AI Readiness Index, 2025.

⁴ Cloud Security Alliance / Google Cloud, "State of AI Security and Governance," December 2025.

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize?

HHS should prioritize one structural change above all others: reorienting its regulatory and reimbursement instruments from behavioral requirements to architectural requirements.

Current frameworks ask behavioral questions — does the AI system meet a performance threshold, does it produce explainable outputs, does the vendor have a responsible AI policy. These are post-hoc assessments of outputs. They do not address the structural question of where clinical authority resides when an AI system is operating between physician and patient.

The evidence for why this reorientation is necessary comes from the federal government's own experience with AI governance. GAO's December 2023 audit of AI implementation across 23 civilian agencies found that all 23 had designated Responsible AI Officials and 20 of 23 had created AI inventories — full behavioral compliance with stated requirements. Yet 15 of the 20 inventories contained incomplete or inaccurate data, two included use cases later determined not to be AI at all, and government-wide guidance for AI acquisition — mandated years earlier — remained unissued.⁵ Agencies achieved compliance. The governance architecture connecting that compliance to accurate outcomes did not function. As of July 2025, nine of GAO's original recommendations remained open.⁶

The specific change: HHS should define a Clinical AI Accountability Architecture standard establishing, for any AI system producing clinically consequential outputs, that three structural conditions must be satisfied before deployment. First, a named responsible party — not a vendor, not a committee — whose accountability is operationally connected to the system's clinical outputs. Second, a measurement specification identifying what metric is being tracked, at what threshold, and at what frequency. Third, a consequence mechanism specifying what is structurally required when the threshold is crossed — not what may happen, but what must happen. For device AI, this standard would operate in parallel with FDA's premarket and postmarket requirements by specifying institutional authority structures — complementing, not duplicating, existing regulatory pathways.

This standard would apply differently across device and non-device AI. For non-medical device AI — the category HHS specifically identifies in Question 3 as raising novel issues — this standard provides the governance specification that existing regulatory frameworks do not supply. It is not a new layer of compliance on top of existing compliance. It is the architectural foundation that makes existing compliance instruments meaningful — the infrastructure that allows current behavioral requirements to connect to operational outcomes. This reorientation represents a category correction in early AI policy — not a repudiation of the compliance work already underway, but the architectural layer that makes it enforceable under operational load. HHS's existing oversight instruments — CMS conditions of participation, ONC certification criteria, accreditation standards — are the substrate on which this architecture would be built, not the systems it replaces.

On reimbursement: CMS payment policy is the fastest market-structuring lever available to HHS. Conditioning reimbursement eligibility for AI-augmented clinical workflows on demonstrated accountability architecture — not on certification of outputs, but on evidence that accountability, measurement, and consequence are structurally present — would restructure vendor incentive architecture within a typical CMS policy cycle. This

could begin as voluntary attestation with phased pilots before mandatory implementation, reducing adoption burden while establishing the architectural standard.

⁵ U.S. Government Accountability Office, "Artificial Intelligence: Agencies Have Begun Implementation but Need to Complete Key Requirements," GAO-24-105980, December 2023. <https://www.gao.gov/products/gao-24-105980>

⁶ U.S. Government Accountability Office, "Artificial Intelligence: Generative AI Use and Management at Federal Agencies," GAO-25-107653, July 2025. <https://www.gao.gov/products/gao-25-107653>

Question 3: What novel legal and implementation issues exist for non-medical device AI, and what role should HHS play?

The novel issue for non-medical device AI is not liability, indemnification, privacy, or security individually. It is what happens when an AI system occupies the clinical decision space between physician and patient without any of those legal frameworks being able to reach where the decisions are actually being made.

The evidence for this is specific and documented. The U.S. Senate Permanent Subcommittee on Investigations found that UnitedHealthcare deployed an AI model (nH Predict) to evaluate post-acute care claims for elderly Medicare Advantage patients. An error rate reported as approximately 90% in internal documents and contemporaneous reporting surrounding the investigation and litigation was associated with decisions that overrode treating physicians' determinations. Denial rates for post-acute care rose from 10.9% to 22.7% between 2020 and 2022. Patients died during or after algorithmic termination of coverage.⁷ These cases predominantly sit on the payer side, but they occupy the clinical decision boundary that patients experience as care — the physician ordered treatment, the algorithm denied it, and the patient bore the consequence.

Separately, a ProPublica investigation documented that Cigna's PXDX algorithm denied over 300,000 payment requests in a two-month period, with reviewing physicians spending an average of 1.2 seconds per claim — a pace consistent with no individualized clinical review. A class action (*Kisting-Leung v. Cigna Corp.*) is proceeding in the Eastern District of California.⁸

In both cases, the AI system operated within existing regulatory frameworks — Medicare Advantage contracts, prior authorization requirements, physician credentialing. The legal

frameworks governing these systems address the payer's contractual obligations, the physician's professional standards, and the patient's appeal rights. None of them address the structural question of where authority resided when the decision was made. The frameworks exist. They were not designed to operate in a space where an AI system is the de facto decision-maker while human authority structures remain nominally intact around it.

This same structural condition extends beyond payer utilization management. Clinical AI systems operating in diagnostic support, sepsis prediction, radiology triage, and treatment recommendation occupy similar decision boundaries — where the system's output shapes clinical action but the accountability architecture was designed for a workflow in which a human clinician was the sole decision agent.

What HHS can do that no existing legal framework can do is define where the authority boundary sits — specifying which clinical outputs require human decision authority as a structural prerequisite, not as an audit finding after the fact. The implementation role HHS should play: develop a Clinical AI Accountability Registry — a federal-level enumeration of clinical AI use cases, the accountability architecture each deploying institution has certified, and the measurement evidence supporting that certification. This does not create new liability. It creates the information infrastructure that makes existing liability frameworks functional in an AI-mediated clinical environment. This registry would be a schema and attestation container — not a centralized patient-data repository — and could be federated through existing reporting rails, building on rather than duplicating existing systems, with voluntary attestation as the default entry point.

⁷ U.S. Senate Permanent Subcommittee on Investigations, "Examining Health Care Denials and Delays in Medicare Advantage," October 2024.

<https://www.hsgac.senate.gov/subcommittees/investigations/hearings/examining-health-care-denials-and-delays-in-medicare-advantage/>

⁸ ProPublica, "How Cigna Saves Millions by Having Its Doctors Reject Claims Without Reading Them." <https://www.propublica.org/article/cigna-pxdx-medical-health-insurance-rejections>. Class action: Kisting-Leung v. Cigna Corp., Eastern District of California.

Question 4: What are the most promising AI evaluation methods, and should HHS support them?

The most consequential gap in current AI evaluation methodology is the gap between pre-deployment testing and post-deployment governance. Most evaluation frameworks treat deployment as the endpoint. Governance requires treating deployment as the starting point.

Pre-deployment evaluation answers: does this system perform acceptably on specified benchmarks? Post-deployment governance answers: is the clinical authority structure surrounding this system still functioning as designed? These are different questions, and the second one is not currently being asked systematically anywhere in the clinical AI evaluation landscape.

The structural concern with pre-deployment testing has deepened. The 2026 International AI Safety Report found that it has become more common for models to distinguish between test settings and real-world deployment, and to exploit loopholes in evaluations — meaning that documented risk of evaluation gaming and test-deploy divergence now exists as a structural condition, not merely a theoretical concern.⁹ This finding applies specifically to the question of why post-deployment governance is necessary — it is not a claim that clinical AI models are routinely or intentionally gaming evaluations, but that the structural gap between test performance and deployment behavior is now documented and must be governed. For any governance framework premised on pre-deployment testing as the primary assurance mechanism, this finding introduces a gap at the foundation: systems that pass evaluation may behave differently under operational conditions, and the evaluation itself cannot detect this divergence.

The specific failure mode this creates in clinical care: a system that performs within evaluated parameters at the model level can simultaneously be producing clinical harm at the institutional level if the human review structures nominally surrounding it have effectively been bypassed by volume, time pressure, or operational design. The evaluation passed. The governance failed. These are not the same event.

HHS should support the development of governance integrity evaluation — a methodology for assessing not whether an AI system produces acceptable outputs but whether the institutional accountability architecture surrounding it remains structurally intact under operational conditions. This is analogous to safety culture audits or quality-of-care audits, but focused on the governance layer rather than the clinical outcome layer. Governance integrity evaluation could be conducted through existing oversight channels — accrediting bodies, internal quality assurance structures, or CMS survey and certification processes — making it extensible without requiring new regulatory authority. It would assess, at defined intervals: whether the named responsible party is operationally connected to what the

system produces, whether the measurement specification is generating data that reaches the people with authority to act on it, and whether the consequence mechanisms specified at deployment are being triggered at the thresholds they were designed for. In a hospital setting, this might mean verifying quarterly that the physician review process for an AI-flagged sepsis alert system is still functioning as designed — that alerts are reaching clinicians, that response times are within specification, and that the escalation pathway activates when the system's alert-to-intervention ratio crosses a defined threshold.

This is evaluable. It requires a different assessment framework than performance benchmarking, but the underlying architecture of what to measure is well-defined and applicable across institution types and AI use cases. The measurement specifications required for governance integrity evaluation are technically feasible with current institutional data infrastructure.

⁹ International AI Safety Report, Bengio, Y. et al., February 2026.

Question 5: What can HHS do to best support private sector activities to promote innovative and effective AI use?

The most valuable thing HHS can do for private sector AI activity in clinical care is create the institutional vocabulary that makes AI governance legible to the decision-makers who control adoption.

Currently, clinical AI governance conversations fail at the institutional level not because the technology is misunderstood but because the governance concepts required to evaluate it don't have stable, shared definitions. GAO documented that six or more different definitions of "AI" existed simultaneously across federal law and guidance — producing definitional instability that propagates through every procurement conversation, every board-level governance discussion, and every regulatory interaction.⁵ Every institution must rebuild foundational shared vocabulary from scratch.

HHS should develop and publish a Clinical AI Governance Lexicon — a standardized set of governance terms with precise institutional definitions, specifying what each term means in an operational context and what evidence satisfies each requirement. This is not a glossary. It is a governance measurement standard — an evidence-binding authority, not merely a definitional one. An institution that can demonstrate its AI system meets the defined standard for "clinical accountability architecture" can move through procurement, accreditation, and regulatory review with measurable confidence. An institution that

cannot has a specific, addressable gap — not a general sense that something might be wrong. Each term in the lexicon maps to required evidence artifacts and escalation triggers — not aspirational principles. The lexicon would be iteratively updated through multi-stakeholder processes, consistent with how ONC and existing standards bodies already operate, ensuring stability without rigidity. Downstream, a lexicon of this kind enables standardized procurement templates, accreditation criteria, and CMS program integrity tools — all built on a common definitional foundation. The lexicon's focus would be on governance-relevant constructs rather than on prescribing technical approaches, ensuring definitional stability does not constrain technical innovation.

The mechanism HHS is best positioned to provide is definitional authority. Private sector accreditation, certification, and testing activities all benefit from having a federal reference standard to work against. Without it, the market produces multiple competing governance frameworks that increase rather than reduce institutional uncertainty. The correlation data supports the value of this approach: organizations with comprehensive governance maturity are nearly twice as likely to achieve successful AI deployment outcomes, suggesting that governance infrastructure and innovation velocity are positively correlated rather than opposed.⁴

⁴ Cloud Security Alliance / Google Cloud, "State of AI Security and Governance," December 2025.

⁵ U.S. Government Accountability Office, GAO-24-105980, December 2023.
<https://www.gao.gov/products/gao-24-105980>

Question 6: Where have AI tools in clinical care met or exceeded expectations, and where have they fallen short?

The pattern of where clinical AI has fallen short is more structurally instructive than any individual case.

AI tools have underperformed consistently where they were deployed into clinical workflows without corresponding governance infrastructure — not governance policy, governance infrastructure. The distinction matters. A payer deploying an AI-driven prior authorization system with a stated commitment to responsible AI deployment, clinical oversight, and appeals processes has governance policy. It has, in the documented cases that have produced the most significant harms, lacked governance architecture: the

structural connections between policy commitments and operational reality that would make those commitments binding rather than aspirational.

Policy says review happens. Architecture makes review impossible to bypass. Where the two diverge — and the evidence shows they have diverged systematically, not occasionally — clinical AI tools fall short not because the model failed but because the governance structure surrounding the model was not designed to hold under operational pressure.

Where clinical AI has met or exceeded expectations, the structural pattern is consistent and identifiable. Constrained deployments with explicit accountability architecture have succeeded: environments where the AI system's scope is bounded, where a named clinical party is responsible for outputs within that scope, where there is a defined measurement specification with a threshold, and where the consequence mechanism — escalation to human review, alert suppression, case flagging — is built into the workflow rather than appended to it. A radiology AI system deployed to flag potential findings for radiologist review, with defined escalation thresholds and explicit tracking of alert-to-review intervals, succeeds because the governance architecture and the clinical workflow are aligned. The AI narrows the decision space; the human retains authority over the decision. The system's accountability infrastructure makes the division legible.

The question HHS should focus on is not which AI tools have worked but what structural conditions made them work. The tools are not the limiting variable. The governance architecture is.

Question 7: What are the administrative hurdles to AI adoption, and how should HHS address them?

The most significant administrative hurdle to clinical AI adoption is not procurement process complexity, documentation burden, or regulatory uncertainty in the abstract. It is the absence of governance infrastructure standards that would allow the institutional legal and risk function to complete its due diligence and approve deployment.

In high-liability clinical AI deployments, the adoption choke point is consistently the legal and risk layer. The General Counsel's office, Chief Risk Officer, and their equivalents in integrated delivery networks and academic medical centers are not opposed to AI. They are unable to approve AI deployments that cannot answer three questions with institutional specificity: who is responsible for what this system does, how will we know if it stops working, and what are we required to do when it doesn't. Where those questions cannot be answered in writing, with named parties and defined thresholds, legal counsel will advise

against deployment — rationally, correctly, and in a way that no amount of clinical performance evidence will override. Clinical medical officers and nursing leadership often raise the initial demand signal for clinical AI; it is the legal and risk function that holds the structural veto.

The enterprise data confirms this pattern. Only 28% of CEOs take direct responsibility for AI governance, and only 17% of boards exercise active oversight of AI governance decisions.¹⁰ Speed-to-market pressure is the top governance barrier — cited by 45% of organizations overall and 56% among technical leaders — creating persistent tension between deployment velocity and the institutional due diligence that legal and risk functions are obligated to perform.² In health systems, this tension is particularly acute because clinical AI outputs carry professional liability implications that general enterprise AI does not.

The administrative hurdles that follow from this are predictable: extended procurement cycles, conditional approvals requiring governance documentation that doesn't exist in standard form, and deployment architectures that add human review requirements without specifying how that review is operationally structured. These are not bureaucratic failures. They are rational institutional responses to governance uncertainty.

The implication for HHS: the fastest path to reducing administrative hurdles to clinical AI adoption is not streamlining the procurement process. It is creating the governance infrastructure standards that allow the legal and risk layer to answer its foundational questions with confidence. When those questions have answers, adoption proceeds. When they don't, process improvements to procurement timelines address the symptom without touching the cause. A US-led Clinical AI Accountability Architecture standard would also position HHS to set the reference point for OECD partners and international AI governance frameworks — translating domestic policy leadership into global standard-setting authority.

¹⁰ McKinsey & Company, "The State of AI," March 2025.

² Pacific AI / Gradient Flow, "2025 AI Governance Survey," December 2025.

Question 8: Where would enhanced interoperability widen market opportunities and accelerate AI development?

The most consequential interoperability gap in clinical AI is not data interoperability. It is governance interoperability — the absence of a common standard for what clinical AI

accountability documentation looks like and how it moves between institutions, regulators, and payers.

Currently, every institution deploying clinical AI produces governance documentation in a locally defined format, against locally defined standards, validated by locally defined processes. When that institution seeks reimbursement, accreditation, or regulatory approval, its governance documentation must be translated into whatever format the receiving institution requires. This translation cost is enormous, largely invisible in discussions of AI adoption barriers, and entirely unnecessary.

The data interoperability work HHS has done through ONC — establishing common data standards and exchange frameworks — demonstrates the model. The same infrastructure logic applies to governance documentation. If HHS established a governance data standard specifying the minimum structured elements of clinical AI accountability documentation — who is responsible, what is being measured, what triggers action — any institution producing compliant documentation would be legible to any other institution in the ecosystem. This standard would reduce, not increase, the documentation burden: vendors could demonstrate compliance with a single, portable governance standard rather than producing bespoke documentation for each institutional buyer. A multi-stakeholder process, building on existing ONC, FDA, and standards-body infrastructure, would govern the standard's development and maintenance.

The market opportunity this unlocks is significant: AI vendors could demonstrate compliance once rather than repeatedly. Payers could incorporate governance verification into prior authorization and coverage decisions using structured, machine-readable attestations. Regulators could evaluate governance architecture across institutions using comparable data. Accreditation bodies could assess governance maturity against a defined standard. The ecosystem gains the interoperability benefit without requiring standardization of AI technology itself.

The concern that a governance data standard creates additional administrative burden is legitimate and addressable. A minimal viable accountability record — a single-page artifact specifying: accountable owner; metric, threshold, and frequency; mandatory escalation action; and evidence link — is structured documentation of the responsible party, measurement specification, and consequence mechanism for each deployed clinical AI system. Institutions already produce this information in fragmented form across procurement records, risk assessments, and compliance documentation. The standard consolidates it; it does not create it. This registry would store governance-artifact metadata

— not patient data — with data minimization, HIPAA alignment, and voluntary attestation as defaults.

Question 9: What challenges do patients and caregivers wish to see addressed, and what concerns do they have?

Patients and caregivers want AI in clinical care to be subject to the same fundamental accountability structure they expect from any clinical decision: someone who is answerable for what it does, a way of knowing whether it is working, and a clear consequence when it isn't.

The specific concern — and the specific unmet need — is accountability displacement. When an AI system is positioned between a patient and the care their physician recommends, and when that system produces a denial or modification without the patient having a legible path to identify who made the decision, why, and with what authority, the patient experiences the same outcome they would from a human clinical error but without the institutional accountability architecture that makes human clinical errors governable. The appeals process assumes a decision-maker. Algorithmic displacement means there may be no decision-maker in the sense the appeals architecture was designed to address.

The documented evidence is consistent with this concern. The AMA survey found that 82% of physicians reported patients abandoning treatment due to AI-driven prior authorization processes.¹ The Senate Permanent Subcommittee on Investigations documented cases where elderly Medicare Advantage patients were denied post-acute care by an AI model with an error rate reported as approximately 90% in internal documents and contemporaneous reporting surrounding the investigation and litigation, and where patients died during or after algorithmic termination of coverage.⁷ In these cases, the patients and their families experienced a clinical outcome shaped by an AI system without a governance pathway that could reach the point where the decision was actually made.

Accountability displacement also raises equity concerns that HHS should address directly. AI systems trained on historical data reproduce patterns present in that data, including disparities in access, treatment, and outcomes across demographic groups.⁹ Where clinical AI operates without accountability architecture that specifies monitoring for distributional outcomes and subgroup impact — with named responsible parties, defined measurement thresholds for disparity detection, and mandatory escalation when those thresholds are crossed — the system can amplify existing disparities invisibly. The proposed standards must structurally encode equity monitoring as an architectural requirement, not an aspirational add-on.

Patients do not need to understand how AI systems work to have this concern satisfied. They need to know that a named institutional party is accountable for clinical AI outputs, that there is a metric telling that party whether the system is working across all patient populations, and that there is a trigger requiring action when it isn't. That is not a technical specification. It is a governance condition — and it is entirely communicable to patients and caregivers in plain language.

What patients are asking for, in structural terms, is the governance architecture that healthcare institutions currently lack the vocabulary to promise and the frameworks to build. HHS's role is to supply both.

¹ American Medical Association, "How AI Is Leading to More Prior Authorization Denials," 2025. <https://www.ama-assn.org/practice-management/prior-authorization/how-ai-leading-more-prior-authorization-denials>

⁷ U.S. Senate Permanent Subcommittee on Investigations, "Examining Health Care Denials and Delays in Medicare Advantage," October 2024. <https://www.hsgac.senate.gov/subcommittees/investigations/hearings/examining-health-care-denials-and-delays-in-medicare-advantage/>

⁹ International AI Safety Report, Bengio, Y. et al., February 2026.

Question 10: What AI research areas should HHS prioritize to accelerate adoption in clinical care?

HHS should prioritize governance infrastructure research — not AI capability research. The limiting variable in clinical AI adoption is not the quality or capability of available AI systems. It is the institutional capacity to deploy those systems within accountability architectures that make them safe to scale.

10a: Published findings on the impact of adopted AI tools in clinical care

The published evidence on clinical AI impact is growing but structurally uneven. Where AI tools have been deployed into governed clinical environments — with defined accountability, measurement specifications, and escalation mechanisms — outcomes data is strong. The COMPOSER sepsis prediction system at UC San Diego Health demonstrated a 17% reduction in sepsis-related mortality across a multi-site implementation, with statistically significant improvements in early detection and intervention timing.¹³ Duke University Health System's Sepsis Watch program produced a

27% reduction in sepsis mortality, with the governance architecture — defined escalation pathways, clinician-in-the-loop review thresholds, and real-time monitoring — identified as a critical factor in sustained performance.

In diagnostic imaging, a 2025 scoping review published in *eClinicalMedicine* analyzed 140 studies of AI in radiology practice and found consistent improvements in diagnostic accuracy and workflow efficiency, including reductions of 20–30 minutes in average read times for certain imaging categories.¹⁴ The review noted, however, that most studies evaluated AI performance in controlled or semi-controlled settings — not under the full operational pressures of clinical deployment.

The structural pattern across this literature is consistent with the governance architecture argument presented throughout this submission: positive outcomes cluster in deployments where accountability infrastructure was designed into the workflow, and failures cluster where AI capability was deployed without corresponding governance architecture. The binding constraint on scaling these successes is not the AI — it is the institutional governance infrastructure surrounding it.

10b: How the literature approaches costs, benefits, and transfers of clinical AI

The cost-benefit literature on clinical AI is methodologically immature in ways that directly affect governance and policy decisions. A 2025 systematic review in *npj Digital Medicine* examined 19 studies of AI cost-effectiveness in healthcare and found that the majority relied on static modeling assumptions that overstated benefits, underreported indirect costs — including workflow disruption, retraining, and governance overhead — and failed to account for cost transfers between institutional actors.¹⁵ A 2025 *medRxiv* systematic review of AI costing methods across 55 studies found that lifecycle costs were consistently omitted, fewer than half of studies reported explicit costing methods, and none addressed the governance infrastructure costs of sustained AI deployment.¹⁶

The structural gap in this literature is not the absence of cost data. It is the absence of a cost framework that accounts for how clinical AI redistributes costs and risks across institutional boundaries. The institution deploying the system bears implementation and governance costs. The vendor captures licensing revenue. The payer captures administrative efficiency gains. The patient bears the clinical risk when governance architecture is absent or degraded. No existing cost-benefit framework in the clinical AI literature models this transfer structure — which means policy decisions based on current cost-effectiveness evidence are operating without visibility into who actually bears the cost when AI governance fails.

HHS-funded research should address this directly by developing cost-benefit methodologies that include governance infrastructure as a required cost category and model the transfer of risk across institutional actors as an explicit analytical dimension.

Research priorities

Specifically, HHS should fund three research programs that do not currently exist in adequate form.

First: clinical AI governance architecture research — systematic study of what accountability structures work in clinical AI deployment, what makes them degrade under operational pressure, and what measurement specifications are most predictive of maintained accountability over time. This is the governance analog to clinical safety research. The field of patient safety research transformed hospital operations by making safety failures measurable and therefore governable. The same transformation is needed for clinical AI governance. This research should produce defined methods and deliverables: governance architecture auditing protocols, degradation indicators, and evidence-based standards for accountability structure design in clinical AI environments. AHRQ, NIH, and CMS demonstration project funding are well-positioned vehicles for this work, and coordination with existing Patient Safety Organization infrastructure would avoid duplication while extending safety science into the AI governance domain.

Second: governance failure taxonomy research — a structured program to catalog the structural conditions that have produced clinical AI harm, analogous to how aviation safety built a systematic incident taxonomy that made systemic failures visible and addressable. Current tracking efforts document the type and pattern of failures — the AI Incident Database cataloged 108 new incidents in its most recent reporting period alone¹² — but these efforts document what happened without systematically analyzing what governance condition was absent. Understanding why an AI system produced harm requires understanding what accountability, measurement, or consequence mechanism was missing — not just what output was wrong. HHS-funded taxonomy research should produce a structured classification of governance failure modes, organized by the specific architectural absence (accountability gap, measurement gap, consequence gap) that allowed harm to occur, enabling pattern detection across incidents and institutions. Coordination with AHRQ's existing Patient Safety Organization network would integrate AI governance failures into the broader patient safety evidence base.

Third: governance interoperability standards development — the research and consensus-building infrastructure to produce the clinical AI governance data standard described in response to Question 8. This is foundational infrastructure for the entire

clinical AI ecosystem and is the kind of public good that the private sector will not produce independently because the value accrues to the ecosystem rather than to any individual firm.

About the Commenter

The Interstitial Institute applies structural governance analysis across public and private sector domains. The frameworks referenced in these comments reflect analysis of AI governance conditions developed through examination of institutional governance architecture across federal AI implementation, healthcare oversight, financial regulation, transportation safety, cybersecurity, and critical infrastructure domains. The Institute's work focuses on identifying and naming the structural conditions between systems where responsibility, measurement, and governance are absent — the spaces where institutional failures persist not because they are unknown but because no governance architecture exists to make them addressable.

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